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Aircraft air intake comprising an outer wall and front and rear frames linked by at least one joining element distinct from the outer wall

Abstract

An aircraft air intake comprising a lip, an internal panel which comprises a front edge linked to the lip, a front frame which comprises an outer edge and an inner edge linked to the lip and/or to the internal panel, a rear frame which comprises an outer edge, directly or indirectly linked to the lip and an inner edge linked to the rear edge of the internal panel, and at least one joining element linking the outer edges of the front frame and of the rear frame.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of French Patent Application Number 2308265 filed on Jul. 31, 2023, the entire disclosure of which is incorporated herein by way of reference.

FIELD OF THE INVENTION

(2) The present application relates to an aircraft air intake comprising an outer wall and front and rear frames linked by at least one joining element distinct from the outer wall and to an aircraft comprising at least one propulsion assembly which comprises such an air intake.

BACKGROUND OF THE INVENTION

(3) According to an embodiment visible in FIG. **1**, an aircraft **10** comprises several propulsion assemblies **12** each comprising an engine system **14** and a nacelle **16** surrounding the engine system **14** and notably making it possible to channel an air flow towards the engine system **14**.

(4) The engine system **14** has an axis of rotation A**14**. Hereinafter in the description, a longitudinal direction is a direction parallel to the axis of rotation **14**. A longitudinal plane is a plane containing

the axis of rotation A**14**. Finally, a transverse plane is a plane at right angles to the axis of rotation A**14**. The front and rear concepts refer to the direction of flow of the air flow in the engine system in operation, which flows from the front to the rear.
(5) As illustrated in FIG. **2**, each nacelle **16** comprises, at the front, an air intake **18** configured to separate a laminar air flow **20** into a laminar internal air flow **20.1** which enters into the nacelle **16** towards the engine system **14** and a laminar external air flow **20.2** flowing out of the nacelle **16**.
(6) According to an embodiment visible in FIGS. **2** and **3**, an air intake **18** comprises a lip **22** which has a C form in cross-section in a longitudinal plane. The lip **22** comprises a leading edge **24.1** which splits the air flow **20** into an internal air flow **20.1** and an external air flow **20.2**, an outer portion **24.2** which extends from the leading edge **24.1** to an outer rear edge **22.1** of the lip **22** over which the external air flow **20.2** flows, and an inner portion **24.3** which extends from the leading edge **24.1** to an inner rear edge **22.2** of the lip **22** and over which the internal air flow **20.1** flows.
(7) The air intake **18** also comprises: an internal panel **26** positioned in the extension of the inner portion **24.3** of the lip **22** which has a front edge **26.1** linked to the inner rear edge **22.2** of the lip **22** and a rear edge **26.2**, a front frame **28**, in ring form, which has a peripheral outer edge **28.1** linked to the outer portion **24.2** of the lip **22** and a peripheral inner edge **28.2** linked to the inner portion **24.3** of the lip **22** and/or to the internal panel **26**, a rear frame **30**, in ring form, which has a peripheral outer edge **30.1** linked to the outer rear edge **22.1** of the lip **22** and a peripheral inner edge **30.2** linked to the rear edge **26.2** of the internal panel **26**.
(8) The front frame **28** and the lip **22** delimit an annular duct **32** in D form, called D-duct.
(9) According to an embodiment, the front and rear frames **28**, **30** are positioned approximately in two transverse planes. In a variant, the front frame can be slightly tapered and form, with a transverse plane, an angle of the order of 10°.
(10) According to one configuration, the internal panel **26** is equipped with an acoustic attenuation structure.
(11) As illustrated in FIGS. **2** and **3**, according to one arrangement, the peripheral outer edge **28.1** of the front frame **28** has a rim **34** pressed against the outer portion **24.2** of the lip **22** and linked thereto by a series of link elements **36** distributed over the entire circumference of the nacelle.
(12) According to another embodiment, the outer rear edge of the lip **22** is linked to the front frame **28** by a first series of link elements. In addition, the air intake comprises an external panel positioned in the extension of the outer portion **24.2** of the lip **22** which has a front edge linked to the front frame and a rear edge linked to the rear frame.
(13) The fact that the lip and the external panel are produced in a single piece as illustrated in FIGS. **2** and **3** makes it possible to not have discontinuity at the surface in contact with the external air flow and therefore to limit the aerodynamic disturbances. However, when the lip extends to the rear frame **30**, its replacement entails dismantling a large number of link elements, namely link elements linking the lip **22**, the internal panel **26** and the peripheral inner edge **28.2** of the front frame **28**, link elements linking the lip **22** and the peripheral outer edge **28.1** of the front frame **28** and finally link elements linking the lip **22** and the rear frame **30**.
(14) According to another drawback, whatever the embodiment, the air intake **18** comprises at least one series of link elements, positioned in line with the front frame **28**, which are flush with a wall of the air intake (the lip **22** and/or an external panel) in contact with the external air flow **20.2**.
(15) In operation, the external air flow **20.2** is laminar from the leading edge **24.1** (also called aerodynamic stagnation point) over a certain length which must be as long as possible to reduce the drag generated by the air intake **18** of the nacelle **16**. Without link elements (like screws, rivets or similar), the theoretical length, characterizing the laminar air zone, could reach a distance of the order of 400 to 950 mm depending on the size of the nacelle. According to the embodiment visible in FIGS. **2** and **3**, the air intake **18** comprises link elements **36** spaced apart from the leading edge **24.1** by a distance less than 500 mm, of the order of 300 mm, which generate disturbances, which is reflected by an increase in the drag of the air intake **18** of the nacelle **16** and, ultimately, in the

energy consumption of the aircraft **10**.

SUMMARY OF THE INVENTION

(16) The present invention aims to remedy all or some of the drawbacks of the prior art.

(17) To this end, the subject of the invention is an aircraft air intake comprising: a lip which comprises a leading edge, an outer rear edge and an inner rear edge, an internal panel which comprises a front edge linked to the inner rear edge of the lip and a rear edge, a front frame which comprises a front peripheral outer edge and a front peripheral inner edge linked to the lip and/or to the internal panel, a rear frame which comprises a rear peripheral outer edge, directly or indirectly linked to the outer rear edge of the lip and a rear peripheral inner edge linked to the rear edge of the internal panel.

(18) According to the invention, the air intake comprises at least one joining element linking the front and rear peripheral outer edges of the front frame and of the rear frame.

(19) Since the front frame is not linked to the lip, the air intake does not have any link element, in line with the front frame, flush with the outer face of the lip. Thus, the air flow which flows in contact with the air intake retains its laminar nature over a greatest possible length.

(20) According to an embodiment, the air intake comprises a single tubular joining element, substantially coaxial to the internal panel.

(21) According to another embodiment, the air intake comprises several joining elements spaced apart from one another, distributed around the internal panel.

(22) According to another feature, each joining element has a front end, the air intake comprising a front link system which links the front end of each joining element and the front peripheral outer edge of the front frame.

(23) According to another feature, the front peripheral outer edge of the front frame is bent back so as to have a substantially cylindrical form. In addition, the front link system comprises at least one series of link elements, distributed over at least one circle, passing through the front peripheral outer edge of the front frame and the front end of each joining element.

(24) According to another feature, each joining element has a rear end, the air intake comprising a rear link system which links the rear end of each joining element and the rear peripheral outer edge of the rear frame.

(25) According to another feature, the rear end of each joining element is bent back and pressed against the rear peripheral outer edge of the rear frame. In addition, the rear link system comprises at least one series of link elements, distributed over at least one circle, passing through the rear end of each joining element and the rear peripheral outer edge of the rear frame.

(26) According to another feature, the lip and each joining element are linked by a link system situated in proximity to the outer rear edge of the lip.

(27) According to another feature, each joining element has a rib, in line with the link system, configured to space apart the lip and each joining element outside of the rib.

(28) According to another feature, for each joining element, the rib has a substantially cylindrical central part, more spaced apart from the internal panel than the rest of the joining element, and two tapered parts linking the central part to the rest of the joining element.

(29) Also a subject of the invention is an aircraft comprising at least one propulsion assembly which comprises an air intake according to one of the preceding features.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features and advantages will emerge from the following description of the invention, a description given purely by way of example, in light of the attached drawings in which:

(2) FIG. **1** is a perspective view of an aircraft and a propulsion assembly,

(3) FIG. **2** is a longitudinal cross-section of an air intake of an aircraft nacelle illustrating an embodiment of the prior art,
(4) FIG. **3** is a longitudinal cross-section of the intake visible in FIG. **2**, the lip being in the dismantled state,
(5) FIG. **4** is a longitudinal cross-section of an air intake of an aircraft nacelle illustrating an embodiment of the invention,
(6) FIG. **5** is a longitudinal cross-section of an air intake of an aircraft nacelle illustrating another embodiment of the invention, the lip being in the dismantled state,
(7) FIG. **6** is a longitudinal cross-section of the air intake visible in FIG. **5**, the lip being in the fitted state,
(8) FIG. **7** is a perspective view of a part of an air intake, without lip, illustrating an embodiment of the invention, and
(9) FIG. **8** is a perspective view of a part of an air intake illustrating an embodiment of the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(10) According to an embodiment, an aircraft comprises at least one propulsion assembly which comprises an air intake **40** to channel an air flow **42** towards an engine system.
(11) Apart from the air intake **40**, the other elements of the propulsion assembly are not represented in FIGS. **4** to **8**. They are known to the person skilled in the art and can be identical to those of the prior art. The air intake **40** has a longitudinal axis coinciding with the engine system axis when the air intake **40** is linked to the engine system.
(12) The air intake **40** comprises a lip **44** which has a C form in cross-section in a longitudinal plane. The lip **44** comprises a leading edge **46.1** which splits the air flow **42** into an external air flow **42.1** and an external air flow **42.2**, an outer portion **46.2** which extends from the leading edge **46.1** to an outer rear edge **44.1** of the lip **44** and over which the external air flow **42.2** flows, as well as an inner portion **46.3** which extends from the leading edge **46.1** to an inner rear edge **44.2** of the lip **44** and over which the internal air flow **42.1** flows.
(13) The lip **44** has an outer face F**44** in contact with the internal and external air flows **42.1**, **42.2** and an inner face F**44**′.
(14) The air intake **40** also comprises: an internal panel **48** positioned in the extension of the inner portion **46.3** of the lip **44**, substantially (+/−10%) coaxial to the longitudinal axis of the air intake, which has a front edge **48.1** linked to the inner rear edge **44.2** of the lip **44** and a rear edge **48.2**, a front frame **50**, in ring form, which has a front peripheral outer edge **50.1** and a front peripheral inner edge **50.2** linked to the inner portion **46.3** of the lip **44** and/or to the internal panel **48**, a rear frame **52**, in ring form, which has a rear peripheral outer edge **52.1** linked to the outer rear edge **44.1** of the lip **44** and a rear peripheral inner edge **52.2** linked to the rear edge **48.2** of the internal panel **48**.
(15) According to an embodiment, the front and rear frames **50**, **52** are positioned approximately in two transverse planes. In a variant, the front or rear frame can be slightly tapered and form, with a transverse plane, an angle of the order of 10° (+\−10%).
(16) According to one configuration, the internal panel **48** is equipped with an acoustic attenuation structure.
(17) The air intake **40** comprises a first link system **54** linking the front edge **48.1** of the internal panel **48** and the inner rear edge **44.2** of the lip **44**. According to one arrangement, the front edge **48.1** of the internal panel **48** is pressed against the inner face F**44**′ of the lip **44** at the inner rear edge **44.2** of the lip **44**. In addition, the first link system **54** comprises a series of link elements **54.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the parts of the lip **44** and of the internal panel **48** which overlap.
(18) According to an embodiment visible in FIG. **4**, the first link system **54** makes it possible to link the front edge **48.1** of the internal panel **48**, the inner rear edge **44.2** of the lip **44** and the front

peripheral inner edge **50.2** of the front frame **50**. In this case, the front peripheral inner edge **50.2** of the front frame **50** is bent back and pressed against the front edge **48.1** of the internal panel **48**. In addition, the first link system **54** comprises a series of link elements **54.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the parts of the lip **44**, of the internal panel **48** and of the front frame **50** which overlap.

(19) According to another embodiment visible in FIG. **6**, the air intake **40** comprises, in addition to the first link system **54**, a second link system **56** linking the front peripheral inner edge **50.2** of the front frame **50** and the internal panel **48**. In this case, the front peripheral inner edge **50.2** of the front frame **50** is bent back and pressed against the internal panel **48** in a zone away from the front edge **48.1** of the internal panel **48**. In addition, the second link system **56** comprises a series of link elements **56.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the front peripheral inner edge **50.2** of the front frame **50** and being housed in the internal panel **48**.

(20) Obviously, the invention is not limited to these embodiments. Thus, the front peripheral inner edge **50.2** of the front frame **50** could be bent back, pressed against the lip **44** and linked thereto in a zone away from the internal panel **48**.

(21) The air intake **40** also comprises a third link system **58** linking the rear peripheral inner edge **52.2** of the rear frame **52** and the rear edge **48.2** of the internal panel **48**. According to one arrangement, the rear peripheral inner edge **52.2** of the rear frame **52** is bent back, pressed against the rear edge **48.2** of the internal panel **48** (or against an element linked to the internal panel **48**) and linked thereto. In addition, the third link system **58** comprises a series of link elements **58.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the rear peripheral inner edge **52.2** of the rear frame **52** and being housed in the internal panel **48**.

(22) According to an embodiment, the air intake **40** comprises a rear connection system **60** configured to link the air intake **40** to the engine system of the propulsion assembly, more particularly to a fan casing of the engine system. According to an embodiment, this rear connection system **60** comprises at least one angle iron which has a first flange **60.1** linked to the internal panel **48** and a second flange **60.2** positioned in a transverse plane and intended to be linked to the fan casing.

(23) According to one configuration, the rear peripheral inner edge **52.2** of the rear frame **52** is bent back, pressed against the first flange **60.1** of the rear connection system **60** which is itself pressed against the internal panel **48**, the link elements **58.1** passing through the rear peripheral inner edge **52.2** of the rear frame **52**, the first flange **60.1** and being housed in the internal panel **48**.

(24) Obviously, the invention is not limited to this embodiment for linking the internal panel **48**, the rear frame **52** and/or the rear connection system **60**.

(25) The air intake **40** comprises a link system **62** directly or indirectly linking the rear peripheral outer edge **52.1** of the rear frame **52** and the outer rear edge **44.1** of the lip **44**.

(26) According to a feature of the invention, the air intake **40** comprises at least one joining element **64** linking the front and rear peripheral outer edges **50.1**, **52.1** of the front frame **50** and of the rear frame **52**.

(27) According to one configuration, the air intake **40** comprises a single tubular joining element **64**, substantially coaxial to the internal panel **48**, which extends over the entire circumference of the air intake **40**. According to one arrangement, the joining element **64** comprises several parts linked to one another to form a single-piece part.

(28) According to another configuration, the air intake **40** comprises several joining elements **64** spaced apart from one another, distributed around the internal panel **48**.

(29) According to an embodiment, each joining element **64** has a front end **64.1** linked to the front frame **50** and a rear end **64.2** linked to the rear frame **52**. In addition, the air intake **40** comprises a front link system **66** linking the front end **64.1** of each joining element **64** and the front peripheral outer edge **50.1** of the front frame **50** and a rear link system **68** linking the rear end **64.2** of each

joining element **64** and the rear peripheral outer edge **52.1** of the rear frame **52**.
(30) According to an embodiment, the front peripheral outer edge **50.1** of the front frame **50** is bent back so as to have a substantially cylindrical form, coaxial to the front end **64.1** of the joining element **64**, and covering the latter. In addition, the front link system **66** comprises at least one series of link elements **66.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the front peripheral outer edge **50.1** of the front frame **50** and the front end **64.1** of each joining element **64**. Obviously, the invention is not limited to this embodiment for the link between the front frame **50** and each joining element **64**.
(31) According to an embodiment, the rear end **64.2** of each joining element **64** is bent back, pressed against the rear peripheral outer edge **52.1** of the rear frame **52**. In addition, the rear link system **68** comprises at least one series of link elements **68.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the rear end **64.2** of each joining element **64** and the rear peripheral outer edge **52.1** of the rear frame **52**. Obviously, the invention is not limited to this embodiment for the link between the rear frame **52** and each joining element **64**.
(32) According to an embodiment, the lip **44** is linked to each joining element **64** by a link system **70**, situated in proximity to the outer rear edge **44.1** of the lip **44**, comprising at least one series of link elements **70.1**, distributed over at least one circle about the longitudinal axis of the air intake **40**, passing through the lip **44** and each joining element **64**.
(33) The joining element **64** is positioned between the lip **44** and the internal panel **48**. Each joining element **64** has, in line with the link system **70**, a rib **72** configured to space apart the lip **44** and each joining element **64** outside of the rib **72**. The rib or ribs **72** is or are positioned in a transverse plane. According to one configuration, for each joining element **64**, the rib **72** has a section in omega form in planes passing through the longitudinal axis of the air intake **40**. Thus, the rib **72** has a central part **72.1** that is substantially cylindrical, more spaced apart from the internal panel **48** and the rest of the joining element **64**, and two tapered parts **72.2**, **72.3** linking the central part **72.1** to the rest of the joining element **64**. Obviously, the invention is not limited to this omega form for the rib **72**. With the joining element **64** being spaced apart from the lip **44** outside of the rib **72**, the latter is configured to press at least one zone of the joining element **64** against the inner face F**44**′ of the lip **44** to link it to the lip by the link system **70**.
(34) Obviously, the invention is not limited to this embodiment for the link linking the lip **44** and the joining element or elements **64**.
(35) According to an embodiment, the different link elements **54.1**, **56.1**, **58.1**, **66.1**, **68.1**, **70.1** can be screws, bolts or rivets.
(36) According to the invention, the front peripheral outer edge **50.1** of the front frame **50** is spaced apart from the lip **44** and is not linked thereto. Consequently, the air intake has no link element, in line with the front frame **50**, flush with the outer face F**44** of the lip **44** in contact with the external air flow **42.2**. Thus, the latter retains its laminar nature over a greatest possible length.
(37) According to another advantage visible in FIGS. **4** and **6**, the lip **44** is linked to the rest of the air intake by only two link systems, which makes it possible to reduce the number of link elements to be dismantled or refitted when dismantling or refitting the lip **44**.
(38) In addition, during dismantling, the rear frame **52** remains in place, thus ensuring the overall stability of the geometry of the air intake.
(39) Finally, the internal panel **48**, the front and rear frames **50**, **52** and the joining element or elements **64** form a rigid box structure which reinforces the air intake **40**, which offers a better absorption of forces in the event of bird strike.
(40) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or

steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. An aircraft air intake comprising: a lip which comprises a leading edge, an outer rear edge, and an inner rear edge, an internal panel which comprises a front edge, linked to the inner rear edge of the lip, and a rear edge, a front frame which comprises a front peripheral outer edge and a front peripheral inner edge linked to the lip, or to the internal panel, or to both, a rear frame which comprises a rear peripheral outer edge directly or indirectly linked to the outer rear edge of the lip and a rear peripheral inner edge linked to the rear edge of the internal panel, and, at least one joining element linking the front peripheral outer edge and the rear peripheral outer edge, wherein the lip and the at least one joining element are linked by a link system situated in proximity to the outer rear edge of the lip, wherein the at least one joining element has a rib, in line with the link system, and configured to space apart the lip and the at least one joining element outside of the rib.
2. The aircraft air intake as claimed in claim 1, wherein the at least one joining element comprises: a single tubular joining element, substantially coaxial to the internal panel.
3. The aircraft air intake as claimed in claim 1, wherein the at least one joining element comprises: several joining elements spaced apart from one another and distributed around the internal panel.
4. The aircraft air intake as claimed in claim 1, wherein the at least one joining element has a front end, and wherein the air intake further comprises a front link system linking the front end of the at least one joining element and the front peripheral outer edge of the front frame.
5. The aircraft air intake as claimed in claim 4, wherein the front peripheral outer edge of the front frame is bent back so as to have a substantially cylindrical form, and wherein the front link system comprises at least one series of link elements, distributed over at least one circle, passing through the front peripheral outer edge of the front frame and the front end of the at least one joining element.
6. The aircraft air intake as claimed in claim 1, wherein the at least one joining element has a rear end and the air intake comprises a rear link system linking the rear end of the at least one joining element and the rear peripheral outer edge of the rear frame.
7. The aircraft air intake as claimed in claim 6, wherein the rear end of each joining element is bent back and pressed against the rear peripheral outer edge of the rear frame, and wherein the rear link system comprises at least one series of link elements, distributed over at least one circle, passing through the rear end of each joining element and the rear peripheral outer edge of the rear frame.
8. The aircraft air intake as claimed in claim 1, wherein the rib has a substantially cylindrical central part, more spaced apart from the internal panel than a rest of the at least one joining element, and two tapered parts linking the substantially cylindrical central part to the rest of the at least one joining element.
9. An aircraft comprising: at least one propulsion assembly which comprises the aircraft air intake as claimed in claim 1.